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Assignment-4

Hill Cipher

1. Cipher Text: HLDTBIHJDXW&CMAQ
Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ Use A=0, B=1, C=2...
Find Plaintext.

→ Find determinant: $|K| = \begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} = 3 \times 7 - 2 \times 5 = 21 - 10 = 11$

$$11^{-1} \bmod 26 = 19$$

$$\begin{aligned} \text{Find inverse: } K^{-1} &= 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26 \\ &= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26 \\ &= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} H &= \begin{bmatrix} 7 \\ 11 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21+154 \\ 33+55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} \bmod 26 \\ &= \begin{bmatrix} 175 \bmod 26 = 19 = T \\ 118 \bmod 26 = 14 = O \end{bmatrix} \end{aligned}$$

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$$\begin{aligned} D &= \begin{bmatrix} 3 \\ 19 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 9+266 \\ 27+95 \end{bmatrix} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \bmod 26 \\ &= \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{matrix} P \\ S \end{matrix} \end{aligned}$$

$$\begin{aligned} O &= \begin{bmatrix} 16 \\ 18 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 18 \end{bmatrix} = \begin{bmatrix} 100 \\ 184 \end{bmatrix} \bmod 26 = \begin{matrix} 4 = E \\ 2 = C \end{matrix} \end{aligned}$$

$$\begin{aligned} H &= \begin{bmatrix} 7 \\ 9 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \bmod 26 = \begin{matrix} 17 = R \\ 4 = E \end{matrix} \end{aligned}$$

$$\begin{aligned} D &= \begin{bmatrix} 3 \\ 23 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \bmod 26 = \begin{matrix} 19 = T \\ 12 = M \end{matrix} \end{aligned}$$

$$\begin{aligned} W &= \begin{bmatrix} 22 \\ 16 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 290 \\ 270 \end{bmatrix} \bmod 26 = \begin{matrix} 14 = E \\ 18 = S \end{matrix} \end{aligned}$$

$$\begin{aligned} C &= \begin{bmatrix} 2 \\ 12 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \bmod 26 = \begin{matrix} 18 = S \\ 0 = A \end{matrix} \end{aligned}$$

$$\begin{aligned} A &= \begin{bmatrix} 0 \\ 6 \end{bmatrix} \quad P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \bmod 26 = \begin{matrix} 6 = G \\ 4 = E \end{matrix} \end{aligned}$$

PT IS TDP SECRET MESSAGE

2. Consider Hill cipher key matrix $\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$

Cipher Text: RXADDY

Find PT for above CT.

→ Determinant: $|K| = \begin{vmatrix} 24 & 19 \\ 5 & 14 \end{vmatrix} = 24 \times 14 - 19 \times 5 = 336 - 95 = 241$

$241 \bmod 26 = 7$
 $7^{-1} = 15$

Inverse Matrix: $K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \bmod 26$
 $K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$

$R = \begin{bmatrix} 17 \\ 23 \end{bmatrix}$ $P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 51 \\ 557 \end{bmatrix} \bmod 26 = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = F$

$A = \begin{bmatrix} 0 \\ 14 \end{bmatrix}$ $P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 27 \end{bmatrix} = D$

$D = \begin{bmatrix} 3 \\ 24 \end{bmatrix}$ $P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 531 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = E$

PT is FLOWER

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3. Confusion Vs Diffusion

Analyze following ciphers w.r.t Confusion & Diffusion:

Cipher	Does it change char value?	Does it change char position?	Confusion/Diffusion	Justification
1. Shift Cipher	Yes	No	Confusion	Each PT char replaced by another char by shifting its value. Position of char is unchanged.
2. Multiplicative Cipher	Yes	No	Confusion	Each char value is multiplied by a key mod 26. The char pos unchanged.
3. Affine Cipher	Yes	No	Confusion	Each char value transformed using $C = (aP + b) \bmod 26$. Pos. of char is unchanged.
4. Transposition Cipher	No	Yes	Diffusion	The char value remain the same, but pos are rearranged acc to the key / trans rule.

4. Transposition Encryption

Encrypt - "The arrow is pointing north direction" using

Simpl: key = 4

Keyword: Keyword: ATLAS

Double: key1 = BREAK

key2 = LOG

→ Simpl: T H E A CT: TRIE9

~~A R D W~~
~~I S D D~~
~~I N I N~~
~~G N O R~~
~~T H D I~~
~~R E R T~~
~~I O N X~~
~~E R E~~
~~E T I~~
~~O N X~~

Simpl: T H E A
A R D W
I S P O
I N T I
N G N O
R T H D
E R E C
T E O N

CT: TRIENRIT
HRSNGTRT EOP
TNHED AWOIODEN

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Keyword: ATLAS

A T L A S
1 5 3 2 4
T H E A R
P O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

CT= TRPERPO
AINNDIX
EWEGHGX
RSTOIEIX
HOONTEN

Double: K1 = BREAK K2 = LOG

~~B B R E A K~~
~~T H E A R~~
~~P O W I S~~
~~P O I N T~~
~~I N G N O~~
~~R T H D I~~
~~R E C T I~~
~~O N X X X~~

~~LOG O G~~
~~A I N~~
~~N D T~~
~~T R P~~
~~R R R~~
~~O E W~~
~~D G H~~
~~E X R~~
~~S T D~~
~~H O D~~
~~N T E~~
~~N X X~~

CT: AINNDIX TRPERPO
EWEGHGX RSTOIEIX HOONTEN

CT: MTPRWHSTUGANTIDICTHNNEDORECROOT

5. Decryption with key 10-Simpli:

i) TIDIE MRNMB REDGO SANOW RSKTP
E E E D O S S S N O A H T U D B I E M P G O D A E P E I X D ELNXX

Key = 10, character = 60 $\rightarrow 60/10 = 6$

T	R	O	O	P	S	H	E	A	I	D
I	N	G	W	E	S	T	N	E	E	
D	M	O	R	E	S	O	P	P	L	
I	E	S	S	E	N	D	G	E	N	
E	R	A	L	D	U	B	O	I	S	
M	E	N	T	O	A	I	D	X	X	

PT = TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS MEN TO AID.

ii) Cipher = ILEDT NHPTR SOIIT MAXSA XERXT
ANONI SNFOT X

keyword: SECRET
Char = $36/6 = 6$

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5	1	4	3	6	
S	E	C	R	E	I
T	H	T	S	I	S
A	P	L	A	I	N
N	I	E	X	T	P
O	R	D	E	M	O
N	S	E	R	A	I
I	O	N	X	X	X

PT = This is the plain
text for demonstration

iii) keyword - PLAIN

Cipher - VEX EAXARA ATXBHGD

$15/3 = 3$

5	3	1	2	4
P	L	A	T	N
H	A	V	E	A
G	R	E	A	T
D	A	Y	X	X

PT = Have a great day.

G. Find Inverse permutation of:

i) $\begin{matrix} 4 & 3 & 1 & 5 & 2 \\ \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \end{matrix}$ $P^{-1} = 3, 5, 2, 4, 1$

ii) $\begin{matrix} 7 & 4 & 1 & 3 & 5 & 2 & 6 \\ \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} & \textcircled{6} & \textcircled{7} \end{matrix}$ $P^{-1} = 3, 6, 4, 2, 5, 7, 1$

iii) $\begin{matrix} 4 & 1 & 3 & 2 \\ \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} \end{matrix}$ $P^{-1} = 2, 4, 3, 1$

iv) $\begin{matrix} 3 & 5 & 1 & 4 & 2 \\ \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \end{matrix}$ $P^{-1} = 3, 5, 1, 4, 2$